out. Other philosophers, including J. J. C. Smart, Hilary Putnam, Karl Popper, Grover Maxwell, Norwood Russell Hanson, and Paul Feyerabend all piled onto the positivists' philosophy of science in the late 1950s and 1960s, building on each other's works and pointing out irreparable flaws in the positivist account of scientific work and progress. Hanson had made many of the same points as Kuhn several years prior to *Structure* in his book *Patterns of Discovery.* (Hanson and Kuhn knew each other and each acknowledged the other's work in their books.) Hanson dubbed the role that nonobservable entities play in scientific work the "theory-laden" practice of science, a name that stuck. This new crop of philosophers largely agreed that the practice of science was theory laden and that the history and practice of actual science was an important guide in developing the philosophy of science. And, while there was much they didn't agree on, a new consensus began to form among professional philosophers of science, a position in opposition to logical positivism, which they called *scientific realism.*

Scientific realism is what it sounds like: the view that there is a real world out there, independent of our observations of it, and that science gives us an approximate description of that world. When a new scientific theory is accepted in place of an old one, this is generally because it gives us a better approximation of the true nature of the world in some important way. This is not to say that the world is entirely insensitive to our probings of it—quantum contextuality assures us that our measurements have some impact on the world—but, by and large, the world proceeds on whether or not we interfere with it. And the contents of that world are approximately described by the content of our best scientific theories, both the observable and unobservable parts.

The realists also argued that the distinction between what is observable and what is unobservable was neither meaningful nor relevant to science. This, of course, was anathema to the positivists. Some positivists had even gone so far as to say that objects seen in microscopes were not truly real, because they were not "directly" perceived. The scientific realists thought this was preposterous. "If this analysis is strictly adhered to, we cannot observe physical things through opera glasses, or even through ordinary spectacles, and one begins to wonder about the status of what we see through an ordinary windowpane," wrote Grover Maxwell, one of the most